ESP-IDF

This chapter contains the following sections. Please read as needed:

- ESP-IDF Getting Started
- Setting Up the Development Environment
- Example

ESP-IDF Getting Started

New to ESP32 ESP-IDF development and looking to get started quickly? We have prepared a general **Getting Started Tutorial** for you.

- Section 1: Environment Setup
- Section 2: Running Examples
- Section 3: Creating a Project
- Section 4: Using Components
- Section 5: Debugging
- Section 6: FreeRTOS
- Section 7: Peripherals
- Section 8: Wi-Fi Programming
- Section 9: BLE Programming

Please Note: This tutorial uses the ESP32-S3-Zero as a teaching example, and all hardware code is based on its pinout. Before you start, it is recommended that you check the pinout of your development board to ensure the pin configuration is correct.

Setting Up the Development Environment

> **INFO**
>
> For the ESP32-S3-Touch-LCD-3.5B development board, ESP-IDF V5.5.0 or above is required.

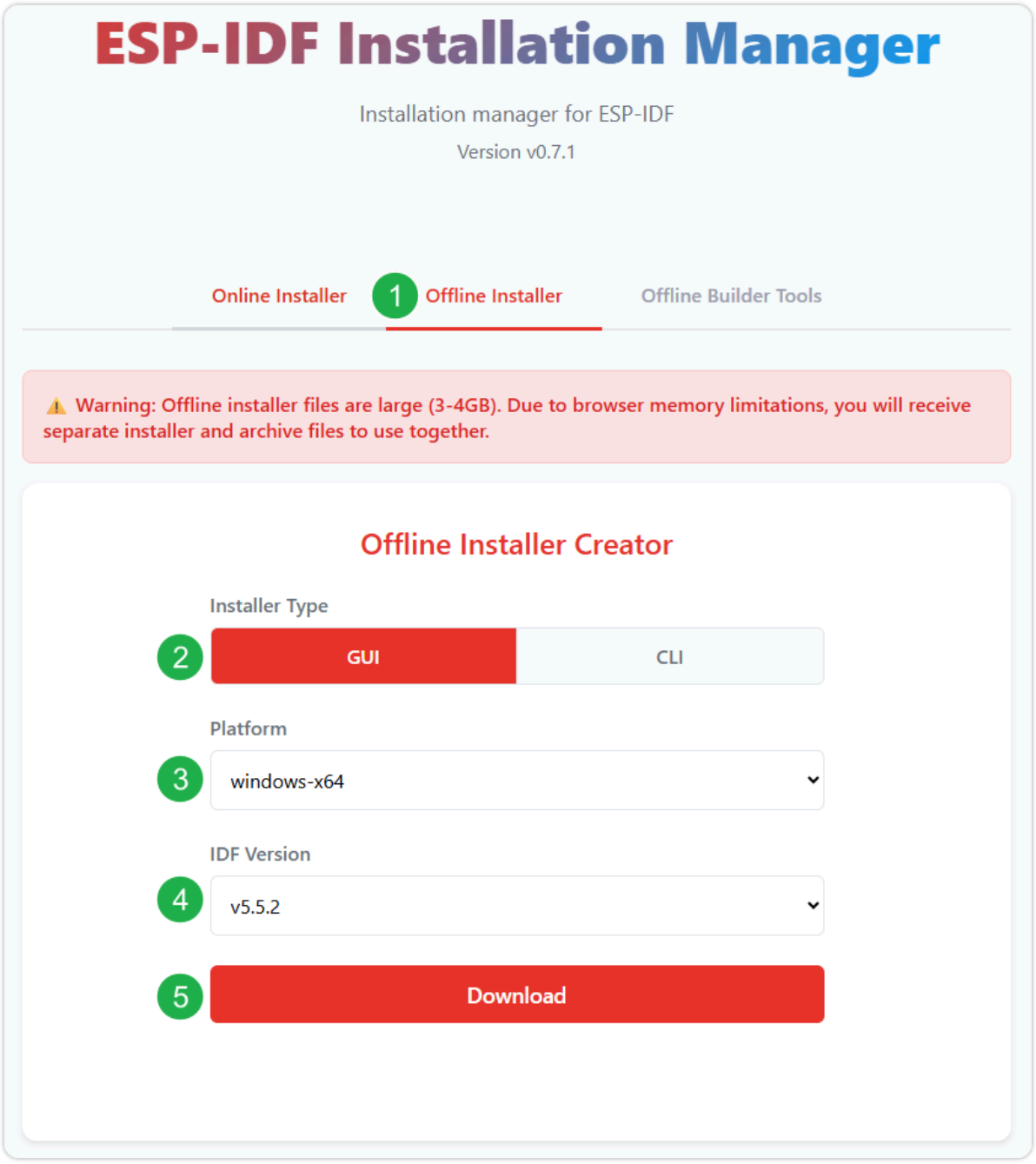
> **NOTE**
>
> The following guide uses **Windows** as an example, demonstrating development using **VS Code + the ESP-IDF extension**. macOS and Linux users should refer to the **official documentation**.

> **VERSION SELECTION**
>
> The screenshots in this section use **ESP-IDF V5.5.2** as an example. When installing, please select the ESP-IDF version that matches your board's example.

Install the ESP-IDF Development Environment

1. Download the installation manager from the ESP-IDF Installation Manager page. This is Espressif's latest cross-platform installer. The following steps demonstrate how to use its offline installation feature.

   Click the **Offline Installer** tab on the page, then select **Windows** as the operating system and **the ESP-IDF version you need** (the version shown in the screenshot is for reference only — choose the version that fits your actual needs).



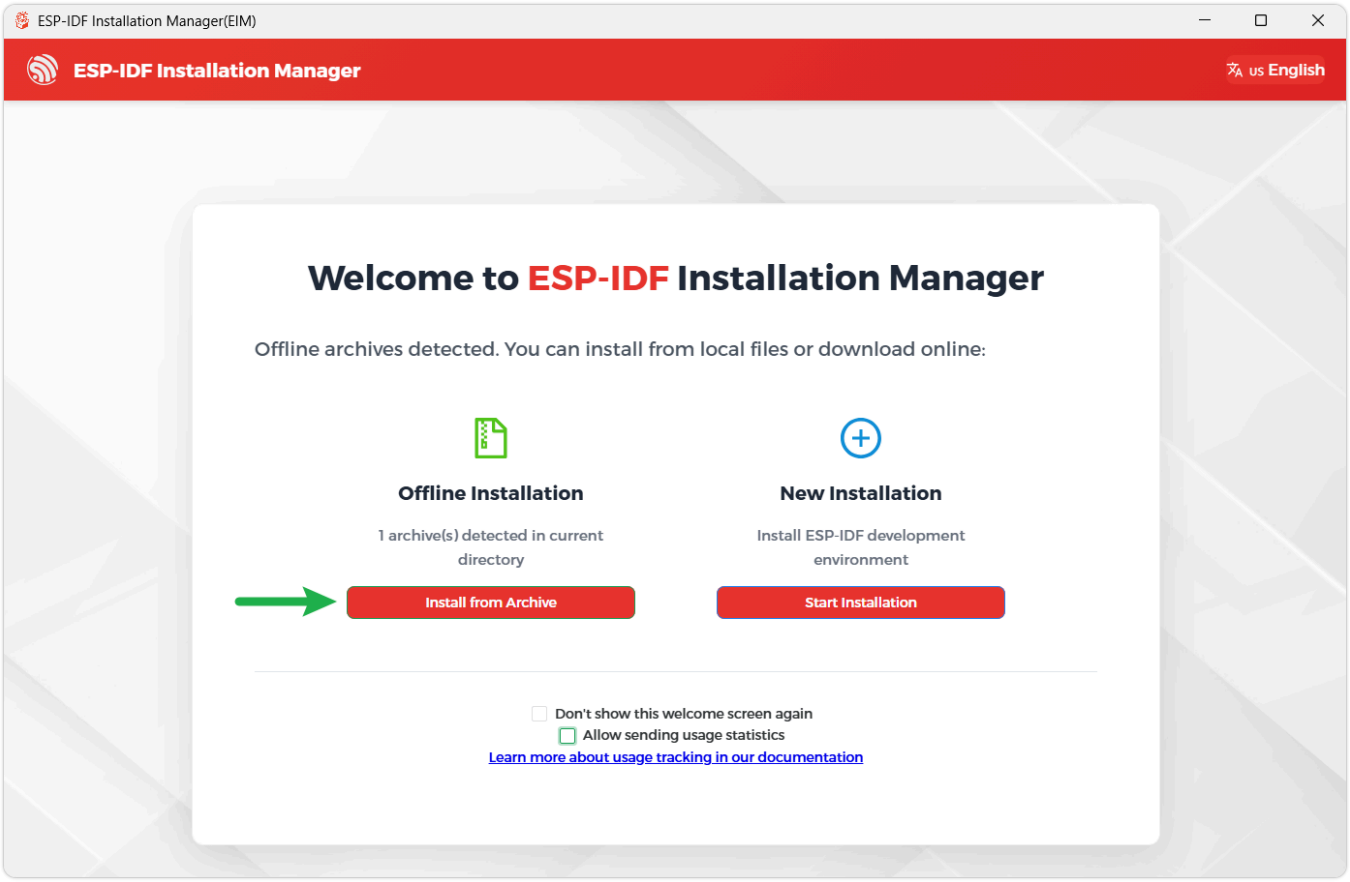
After confirming your selection, click the download button. The browser will automatically download two files: the **ESP-IDF Offline Package (.zst)** and the **ESP-IDF Installer (.exe)**.



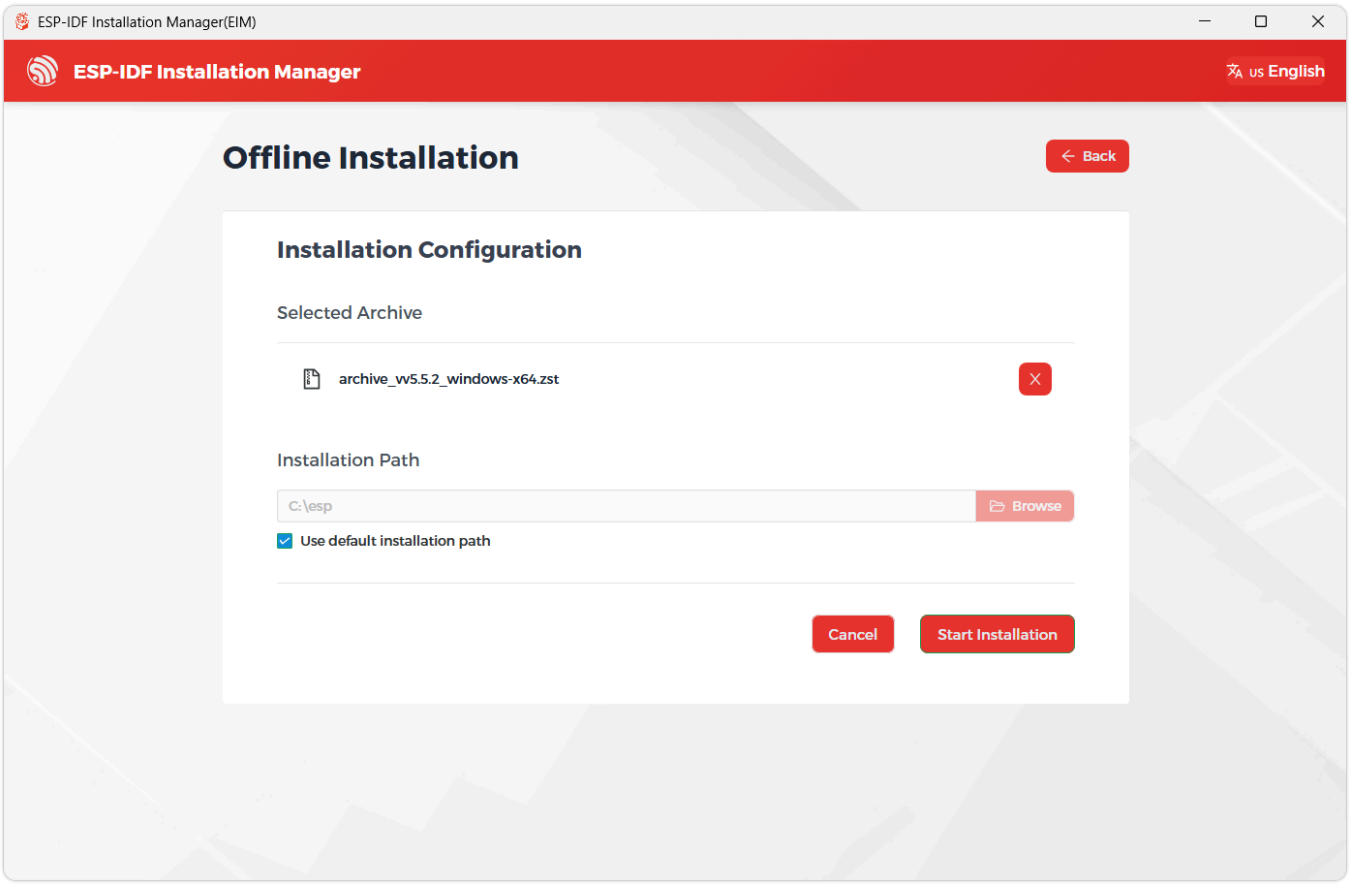
Please wait for both files to finish downloading.

2. Once the download is complete, double-click to run the **ESP-IDF Installer (eim-gui-windows-x64.exe)**.

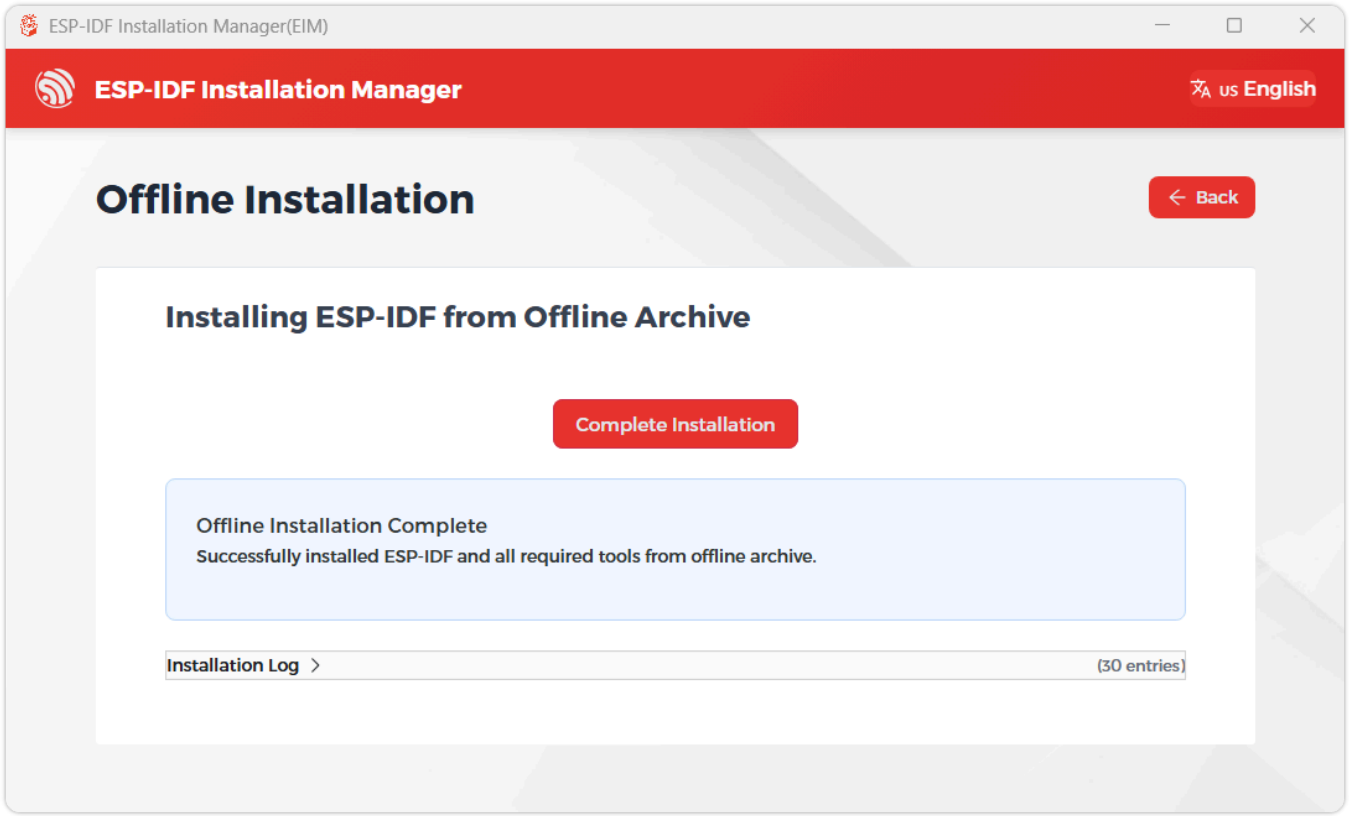
   The installer will automatically detect if the offline package exists in the same directory. Click **Install from archive**.



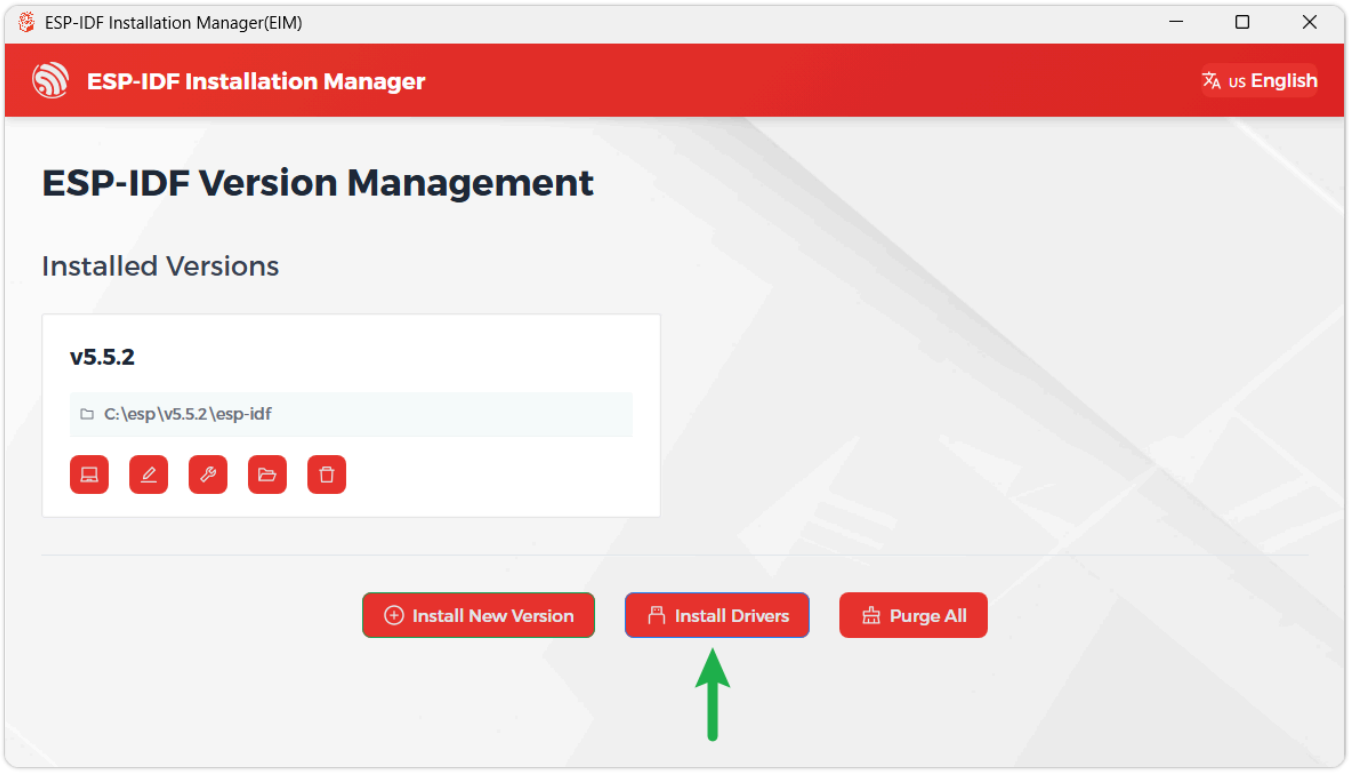
Next, select the installation path. We recommend using the default path. If you need to customize it, ensure the path does not contain Chinese characters or spaces. Click **Start installation** to proceed.



3. When you see the following screen, the ESP-IDF installation is successful.



4. We recommend installing the drivers as well. Click **Finish installation**, then select **Install driver**.



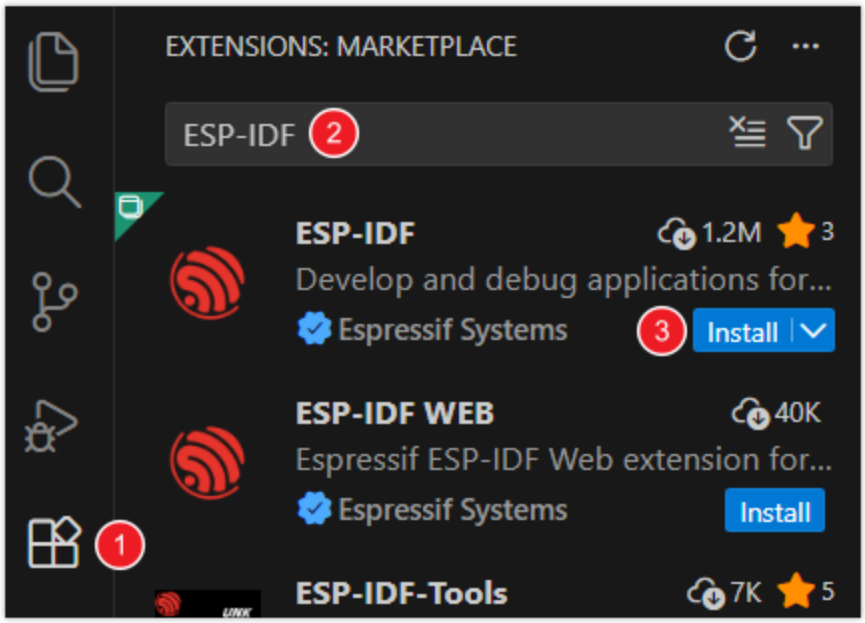
Install Visual Studio Code and the ESP-IDF Extension

1. Download and install Visual Studio Code.

2. During installation, it is recommended to check **Add "Open with Code" action to Windows Explorer file context menu** to facilitate opening project folders quickly.

3. In VS Code, click the **Extensions** icon  in the Activity Bar on the side (or use the shortcut `Ctrl` + `Shift` + `X`) to open the **Extensions** view.

4. Enter **ESP-IDF** in the search box, locate the ESP-IDF extension, and click Install.



5. For **ESP-IDF extension versions ≥ 2.0**, the extension will automatically detect and recognize the ESP-IDF environment installed in the previous steps, requiring no manual configuration.

Example

The ESP-IDF examples are located in the `ESP-IDF` directory of the example package.

Example	Basic Program Description	Dependency Library
01_factory	Comprehensive test program	-
02_qmi8658_example	Print IMU data	-
03_pcf85063_example	Print data from the RTC	-
04_axp2101_example	Print data from the power management IC	-
05_lvgl_example	Run the official LVGL Demo UI	-
06_lvgl_image	Display an image using LVGL	-
07_lvgl_image_sd	Display an image from the TF card using LVGL	-

01_factory

Example Description

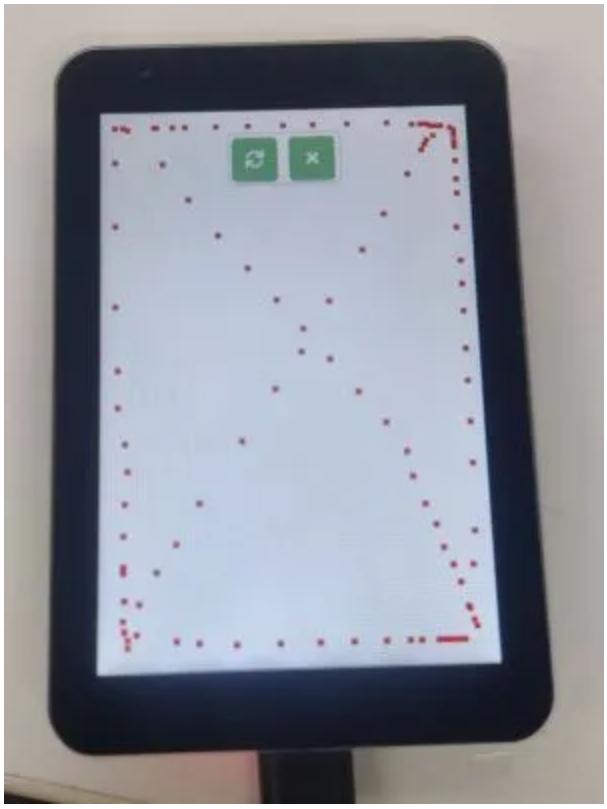
- This is a comprehensive example for the ESP32-S3-Touch-LCD-3.5B, and is also the factory-default flashed example.

Hardware Connection

- Connect the board to the computer.

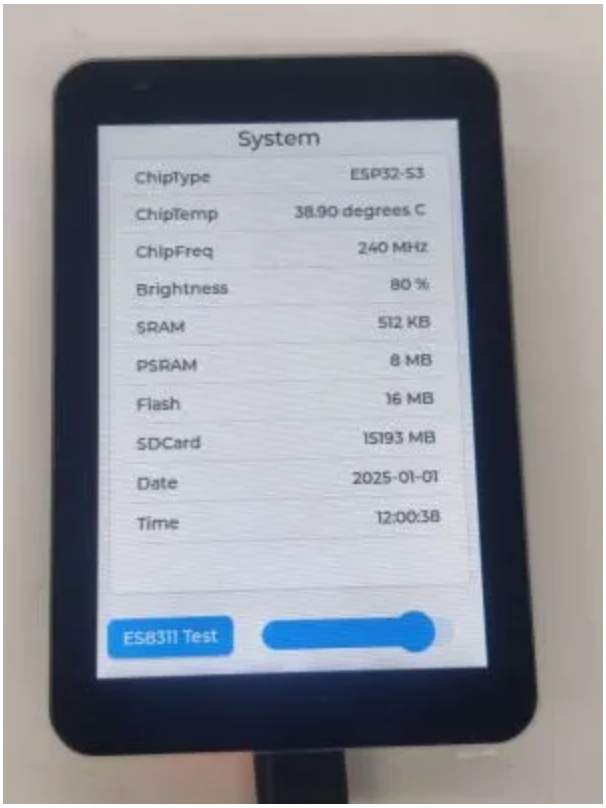
Operation Result

- Touch test: slide your finger on the screen to see the trajectory. Press "X" to exit the touch test.

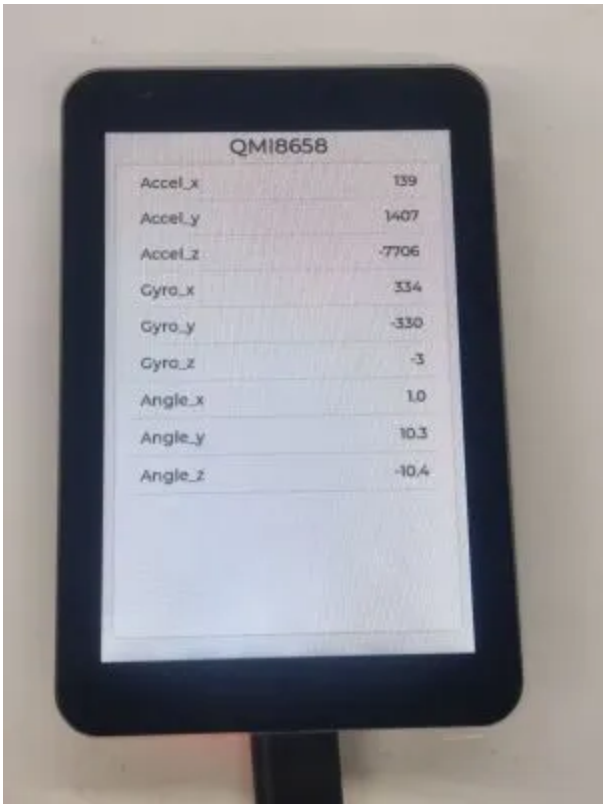


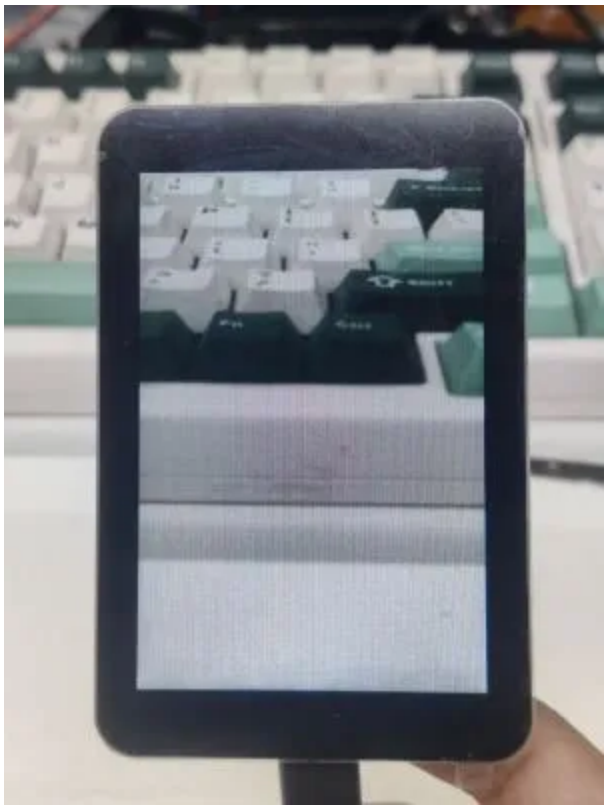
- Swipe left/right to switch between screens.

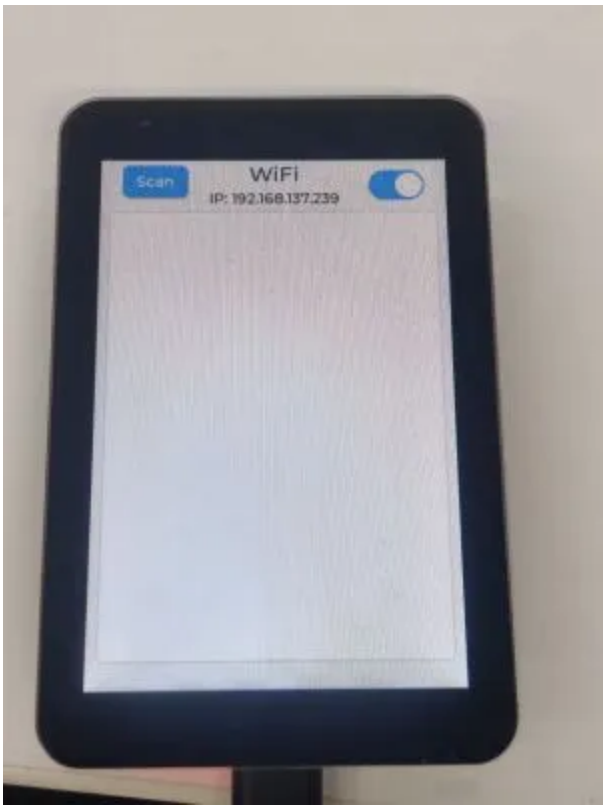



System
ChipType ESP32-S3
ChipTemp 38.90 degrees C
ChipFreq 240 MHz
Brightness 80 %
SRAM 512 KB
PSRAM 8 MB
Flash 16 MB
SDCard 15193 MB
Date 2025-01-01
Time 12:00:38
ES8311 Test


AXP2101
isCharging NO
isBatteryConnect NO
isVbusIn YES
BatteryPercent -1 %
BatteryVoltage 0 mV
VbusVoltage 4948 mV
SystemVoltage 4952 mV
DC1Voltage 3300 mV
ALDO1Voltage 3300 mV
BLDO1Voltage 1500 mV
BLDO2Voltage 2800 mV


QMI8658
Accel_x 139
Accel_y 1407
Accel_z -7706
Gyro_x 334
Gyro_y -330
Gyro_z -3
Angle_x 1.0
Angle_y 10.3
Angle_z -10.4





02_qmi8658_example

Example Description

- This example demonstrates driving the QMI8658 on the ESP32-S3-Touch-LCD-3.5B to read and print accelerometer, gyroscope, and IMU temperature data.

Hardware Connection

- Connect the board to the computer.

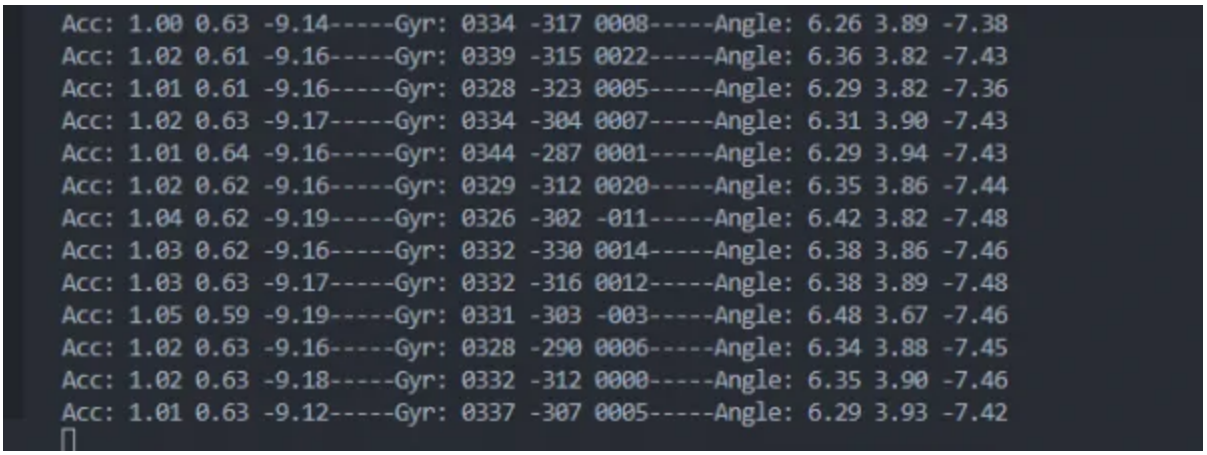
Code Analysis

- Initialize and start the test program

```
i2c_master_bus_handle_t i2c_bus_handle = bsp_i2c_init();
bsp_qmi8658_init(i2c_bus_handle);
bsp_qmi8658_test();
```

Operation Result

- The screen shows no output
- Open the serial monitor



03_pcf85063_example

Example Description

- This example demonstrates driving the PCF85063 on the ESP32-S3-Touch-LCD-3.5B to set the time/date and read the time.

Hardware Connection

- Connect the board to the computer.

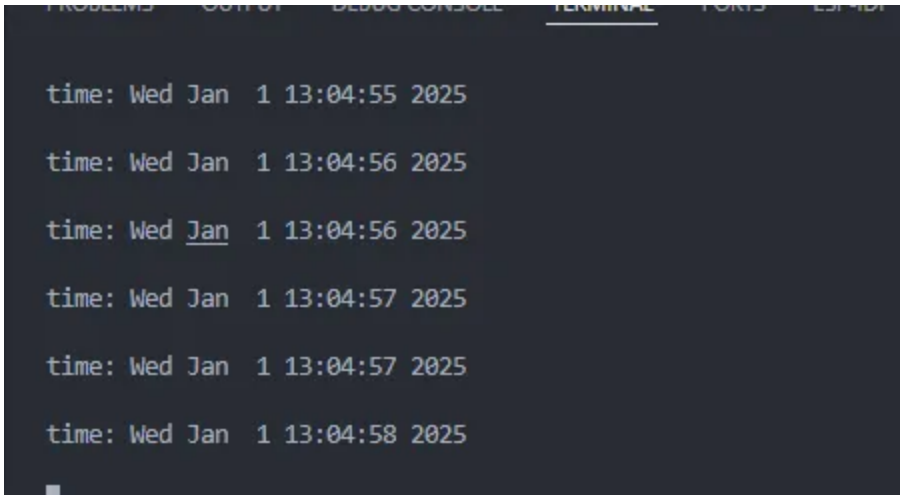
Code Analysis

- Initialize and start the test program

```
i2c_master_bus_handle_t i2c_bus_handle = bsp_i2c_init();
bsp_pcf85063_init(i2c_bus_handle);
bsp_pcf85063_test();
```

Operation Result

- The screen shows no output
- Open the serial monitor



04_axp2101_example

Example Description

- This example demonstrates using XPowers to drive the AXP2101 and printing data via the serial port.

Hardware Connection

- Connect the board to the computer.

Code Analysis

- Initialize

```
i2c_master_bus_handle_t i2c_bus_handle = bsp_i2c_init();
bsp_axp2101_init(i2c_bus_handle);
```

Operation Result

- The screen shows no output
- Open the serial monitor

```
DC2  : +   Voltage:1000 mV
DC3  : +   Voltage:3300 mV
DC4  : +   Voltage:1000 mV
DC5  : +   Voltage:3300 mV
DCDC=================================================================
DC1  : +   Voltage:3300 mV
DC2  : +   Voltage:1000 mV
DC3  : +   Voltage:3300 mV
DC4  : +   Voltage:1000 mV
DC5  : +   Voltage:3300 mV
ALDO=================================================================
ALDO1: +   Voltage:3300 mV
ALDO2: +   Voltage:3300 mV
ALDO3: +   Voltage:3300 mV
ALDO4: +   Voltage:3300 mV
BLDO=================================================================
BLDO1: +   Voltage:1500 mV
BLDO2: +   Voltage:2800 mV
CPUSLDO==============================================================
CPUSLDO: + Voltage:1000 mV
DLDO=================================================================
DLDO1: +   Voltage:3300 mV
DLDO2: +   Voltage:3300 mV
=====================================================================
PowerKeyPressOffTime:4 Second
PowerKeyPressOnTime:128 Ms
=====================================================================
getDCHighVoltagePowerDownEn:ENABLE
getDC1LowVoltagePowerDownEn:ENABLE
getDC2LowVoltagePowerDownEn:ENABLE
getDC3LowVoltagePowerDownEn:ENABLE
getDC4LowVoltagePowerDownEn:ENABLE
getDC5LowVoltagePowerDownEn:ENABLE
I (938) main_task: Returned from app_main()
```

05_lvgl_example

Example Description

- This example demonstrates running the LVGL example program on the ESP32-S3-Touch-LCD-3.5B.

Hardware Connection

- Connect the board to the computer.

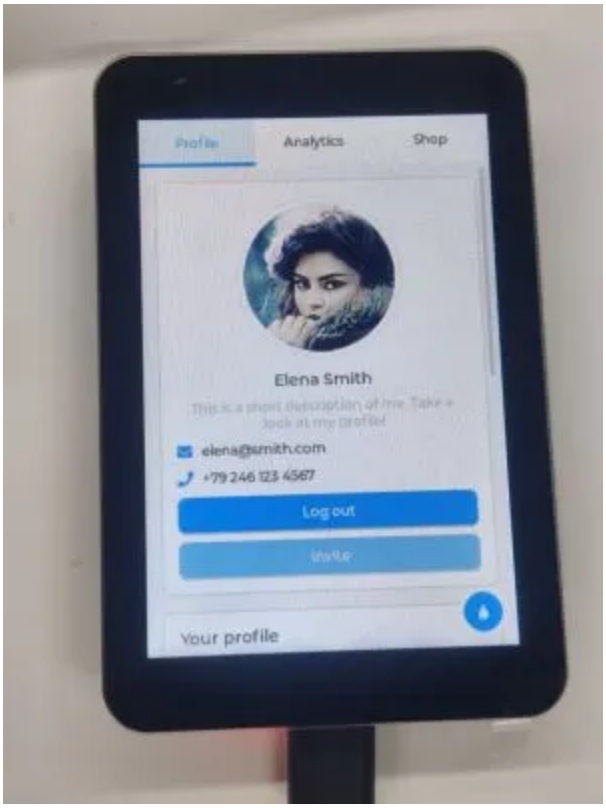
Code Analysis

- Select the LVGL example to run.

```
// lv_demo_benchmark();
// lv_demo_music();
lv_demo_widgets();
```

Operation Result

- Touch operation is supported.



06_lvgl_image

Example Description

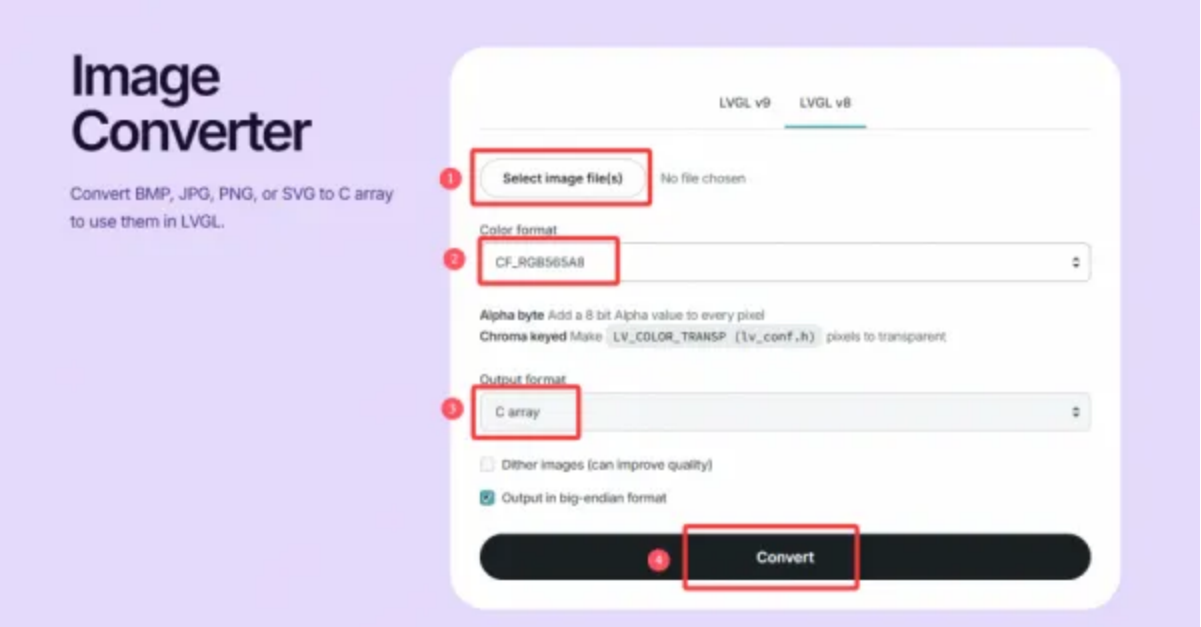
- This example demonstrates displaying an image with LVGL on the ESP32-S3-Touch-LCD-3.5B.

Hardware Connection

- Connect the board to the computer.

Preparation

- Open LVGL image converter website.
- Import the image `images/image_1.jpg` and convert it; this will generate an `image_1.c` file.



- Copy `image_1.c` to the `main` folder.
- Add `image_1.c` to the `CMakeLists.txt` in the `main` folder

```
idf_component_register(SRCS "main.cpp" "image_1.c"INCLUDE_DIRS ".")
```

Code Analysis

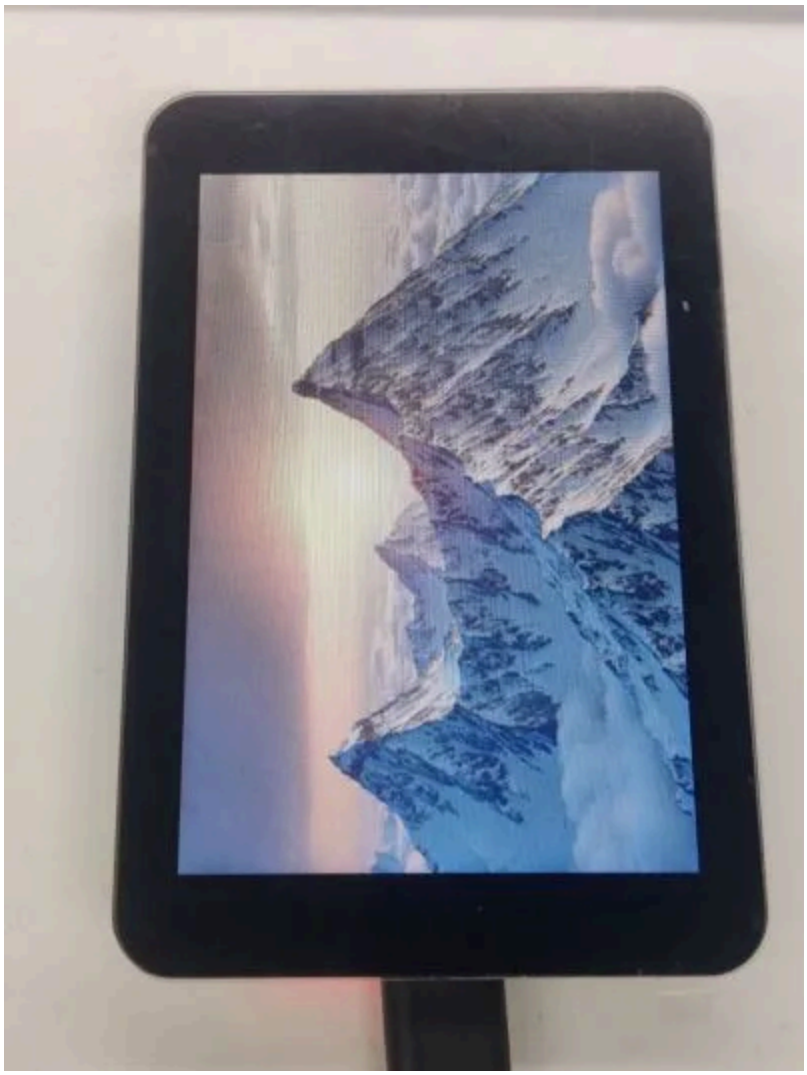
- In `main.c`, add the following code to declare the image.

```
LV_IMG_DECLARE(image_1);
```

- Specify the image to be displayed.

```
lv_img_set_src(img, &image_1);
```

Operation Result



07_lvgl_image_sd

Example Description

- This example demonstrates displaying images from the TF card using LVGL on the ESP32-S3-Touch-LCD-3.5B, supporting formats such as jpg, bmp, png, bin, etc.

Hardware Connection

- Connect the board to the computer.
- Insert the TF card into the card slot.

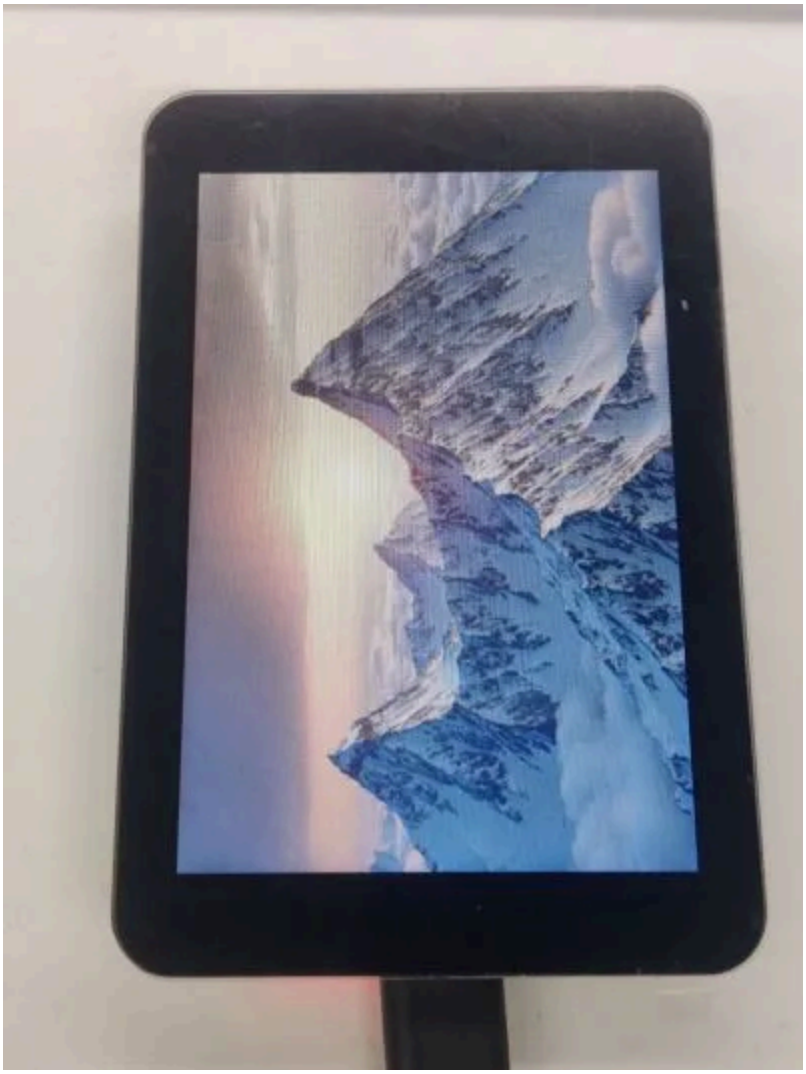
Preparation

- Insert the TF card into the computer.
- Copy the images folder of this project to the root directory of the TF card.

- Insert the TF card into ESP32-S3-Touch-LCD-3.5B

Operation Result

- Touch operation is supported.



Give Feedback